

CS 736: Medical Image Computing

(Instructor : Suyash P. Awate)

End-Semester Examination

Date: 6 May 2021

Duration: 3 hours

Roll Number: _____

Name: _____

When in doubt, make suitable assumptions, state them clearly, and proceed to solve the problem.
All the best!

Formulae for Reference

Poisson probability mass function: $P(x; \lambda) := \lambda^x \exp(-\lambda)/x!$ with mean = λ , variance = λ , and mode = $\text{floor}(\lambda)$.

Rician probability density function: $P(x; \nu, \sigma) := \frac{x}{\sigma^2} \exp(-\frac{x^2 + \nu^2}{2\sigma^2}) I_0(\frac{x\nu}{\sigma^2}), \forall x \geq 0$

Gamma probability mass function: $P(x; \alpha, \beta) := \eta(\alpha, \beta) x^{\alpha-1} \exp(-\beta x)$, where $\eta(\alpha, \beta)$ is a normalization constant that is a function of parameters α and β . In this case, mean = α/β , variance = α/β^2 , and mode = $(\alpha - 1)/\beta$.

Rayleigh probability density function: $P(x; \sigma) := \frac{x}{\sigma^2} \exp(-\frac{x^2}{2\sigma^2}), \forall x \geq 0$

Matrix M is skew-symmetric iff $M^\top = -M$

If X is a real-valued $N \times N$ matrix, then the matrix exponential $\exp(X)$ equals $\sum_{k=0}^{\infty} X^k / (k!)$

Every special orthogonal matrix can be represented as a matrix exponential of a skew-symmetric matrix.

1. [25 points] Consider a set of N MRI images $\{R_n(\cdot)\}_{n=1}^N$ with associated binary segmentations of an object of interest indicated by the images $\{S_n(\cdot)\}_{n=1}^N$. Thus, at voxel location x , the MRI-image intensity is $R_n(x)$ and the binary label is $S_n(x)$.
 - (10 points: 6 + 4) Formulate an optimization problem to find (i) a mean template image for the MRI, say, $R(\cdot)$, and (ii) a mean template image for the segmentation, say, $S(\cdot)$. Assume affine transformations within registration. Also, describe an implementable algorithm.
 - (5 points) Suppose you want to use the mean template MRI image $R(\cdot)$ and the mean template segmentation image $S(\cdot)$ to segment a new MRI image $Q(\cdot)$. Formulate any optimization problems required to be solved. Also, describe an implementable algorithm.
 - (10 points) Suppose you want to segment a new image $T(\cdot)$ without using the mean templates, but using directly the original dataset $\{(R_n(\cdot), S_n(\cdot))\}_{n=1}^N$. Formulate any optimization problems required to be solved. Also, describe an implementable algorithm.
2. [35 points]

Consider a set of points $\{x_n \in \mathbb{R}^D\}_{n=1}^N$ that are implicitly mapped into a reproducing-kernel Hilbert space (RKHS), with a kernel function $k(\cdot, \cdot)$, as $\{k(x_n, \cdot)\}_{n=1}^N$.

 - (10 points: 2 + 4 + 4) Describe an implementable algorithm to perform principal component analysis in the mapped space to get (i) the mean, (ii) the modes of variation, and (iii) the variances along the modes of variation. How is a mode of variation represented in the RKHS ?
 - (10 points) Describe an implementable algorithm to represent the dataset into a space of reduced dimension $E \ll D$ by projecting each mapped point $k(x_n, \cdot)$ in RKHS onto the subspace represented by the top (most prominent) K modes of variation.
 - (10 points) Assume $k^1(\cdot, \cdot)$ and $k^2(\cdot, \cdot)$ are kernel functions. Consider the following statement: $k^3(\cdot, \cdot)$ defined as $k^3(x, y) := k^1(x, y) + k^2(x, y) + k^1(x, y)k^2(x, y)$ is also a kernel function. Is this statement true or false ? Argue logically based on the concepts covered in class.
 - (5 points: 2 + 3) When using a radial-basis-function as the kernel function, what structure does the kernel impart to the distribution of the implicitly mapped data in RKHS ? Argue logically based on the concepts covered in class. State at least two distinct properties of this structure.
3. [40 points] Consider a collection of observed binary images $\{y_n(\cdot)\}_{n=1}^N$ of an anatomical structure of interest within a population of subjects. Thus, each pixel in the image takes intensities 0 or 1. Consider the images $\{y_n(\cdot)\}_{n=1}^N$ to be already registered to a common anatomical space.

Suppose you know that this set of N images comprises K clusters, modeled by a set of cluster-representative images $\{\mu_k(\cdot)\}_{k=1}^K$ that contain real values within $(0, 1)$.

Suppose you want to estimate the cluster-representative images (modeled as parameters), while modeling the cluster label, say, z_n , associated with each observed image $y_n(\cdot)$ as a hidden/latent random variable. You decide to use the expectation-maximization (EM) optimization framework for the inference.

 - (6 points) Formulate all necessary statistical models within the $Q(\cdot)$ function within the EM framework. Your models must respect the state spaces associated with all variables involved. Specify all necessary constraints on the variables underlying the statistical models.
 - (4 points) Specify the formulae for the “membership” values within the E step.
 - (10 points) Clearly state an implementable algorithm for performing the optimization of all parameters within the M step.

- (8 points) Suppose you want to incorporate an edge-preserving smoothness/regularity prior on all the cluster-representative images $\{\mu_k(\cdot)\}_{k=1}^K$. Specify this prior model exactly.
 - (12 points) Specify how the EM optimization algorithm can be modified in the previous sub-question to accommodate this prior model.
4. [20 points] Consider a 3D object of interest whose boundary is represented by a set of well-defined N landmark points, i.e., $\{u_n\}_{n=1}^N$. In the process of measuring these landmark locations, there are inevitable errors that introduce independent and identically distributed perturbations following standard Laplace distributions over each of the three physical dimensions.
- A group of three people independently measure the pointset. However, each employ a different physical coordinate frame, with different origins, axes, and scales. Furthermore, each person is able to measure only a subset of the landmarks. The first person measures $I < N$ landmarks, the second person measures $J < N$ landmarks, and the last person measures $K < N$ landmarks. Between the three people, we know that each landmark gets measured at least once.
- (8 points) Formulate an optimization problem to estimate the shape represented by underlying uncorrupted pointset, say, $\{u_n\}_{n=1}^N$, given the three measured corrupted pointsets.
 - (12 points) Describe an implementable algorithm to perform the optimization.